

Information Retrieval #10 — Probabilistic Retrieval

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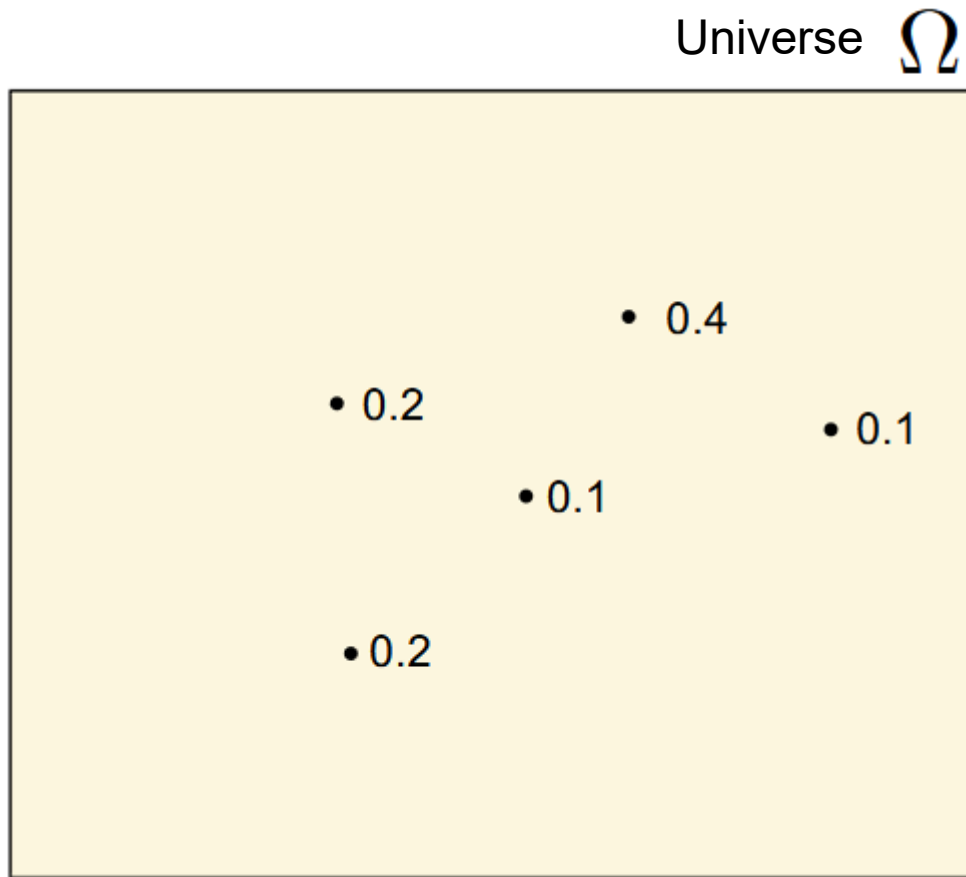
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1. Probabilistic Retrieval
2. Language Models
3. Kahoot

Probability Theory



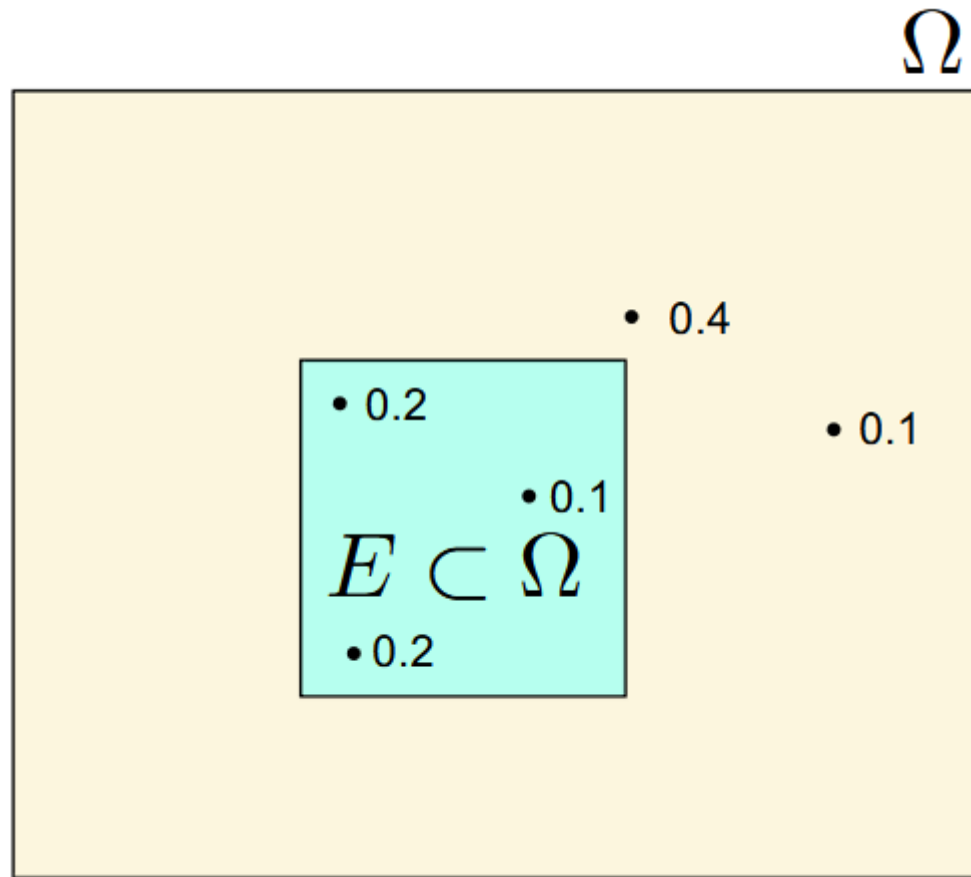
Probability Distribution

$$P : \Omega \rightarrow [0, 1]$$
$$\omega \mapsto P(\omega)$$

Elementary Event

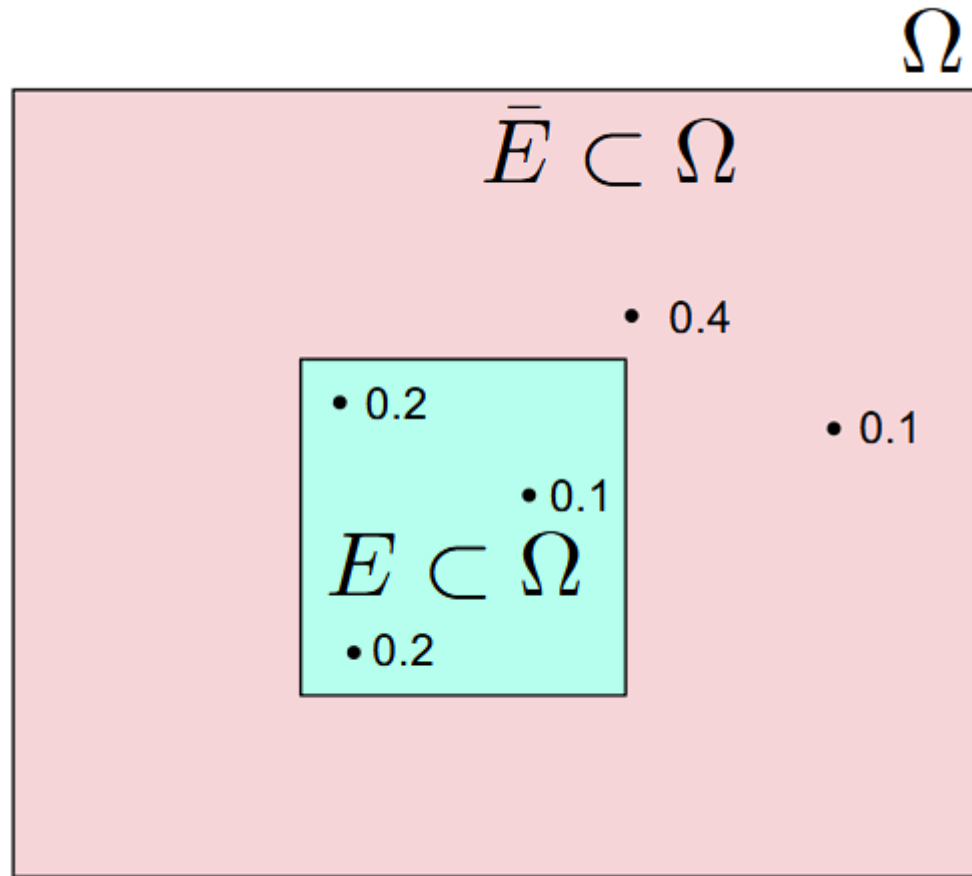
$$\sum_{\omega \in \Omega} P(\omega) = 1$$

Event



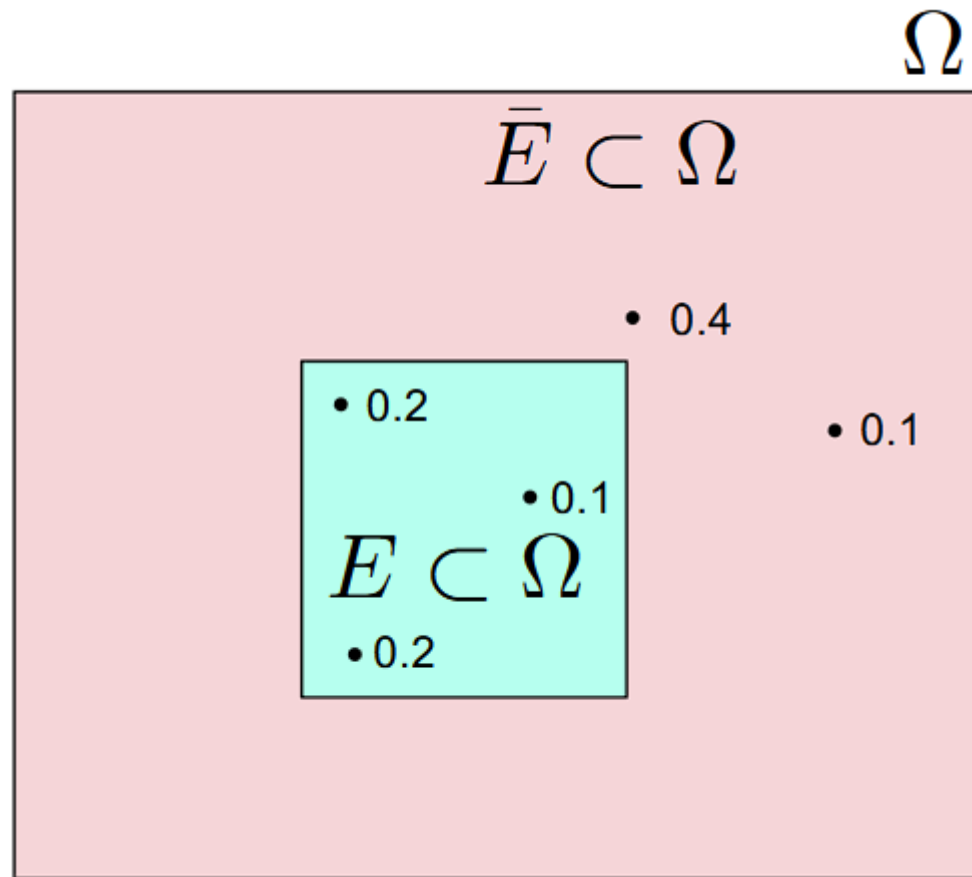
$$P(E) = \sum_{\omega \in E} P(\omega) \in [0, 1]$$

Complement



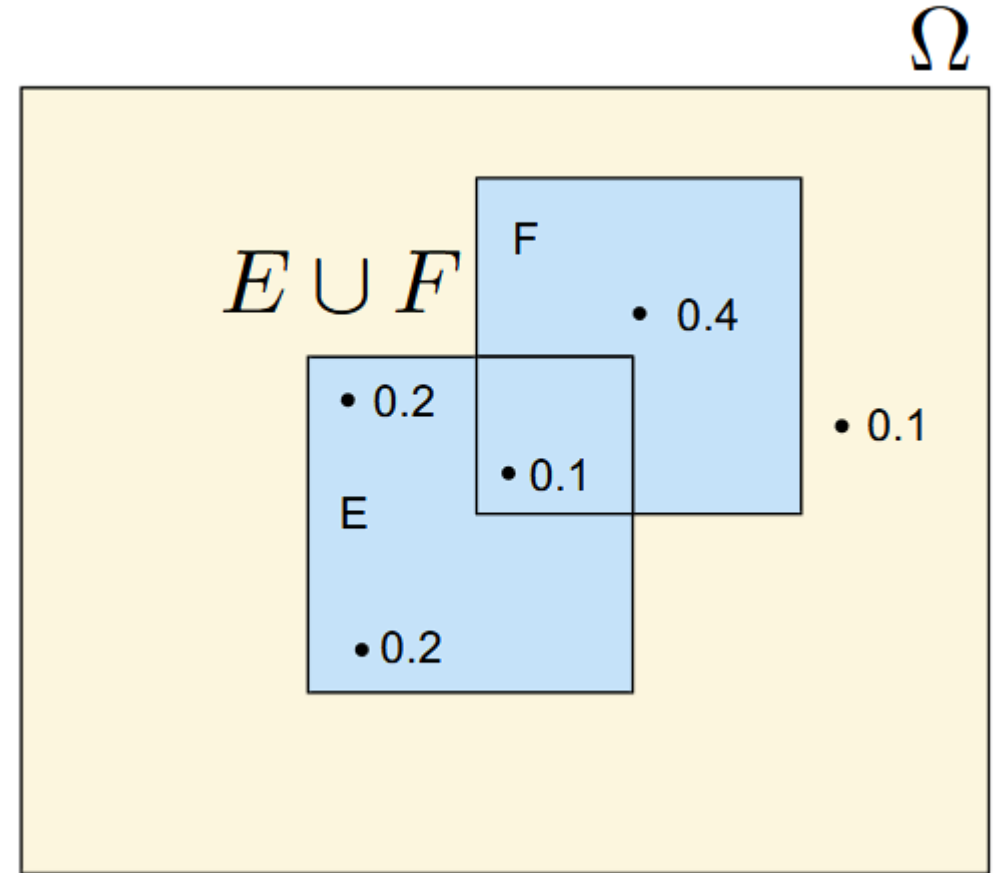
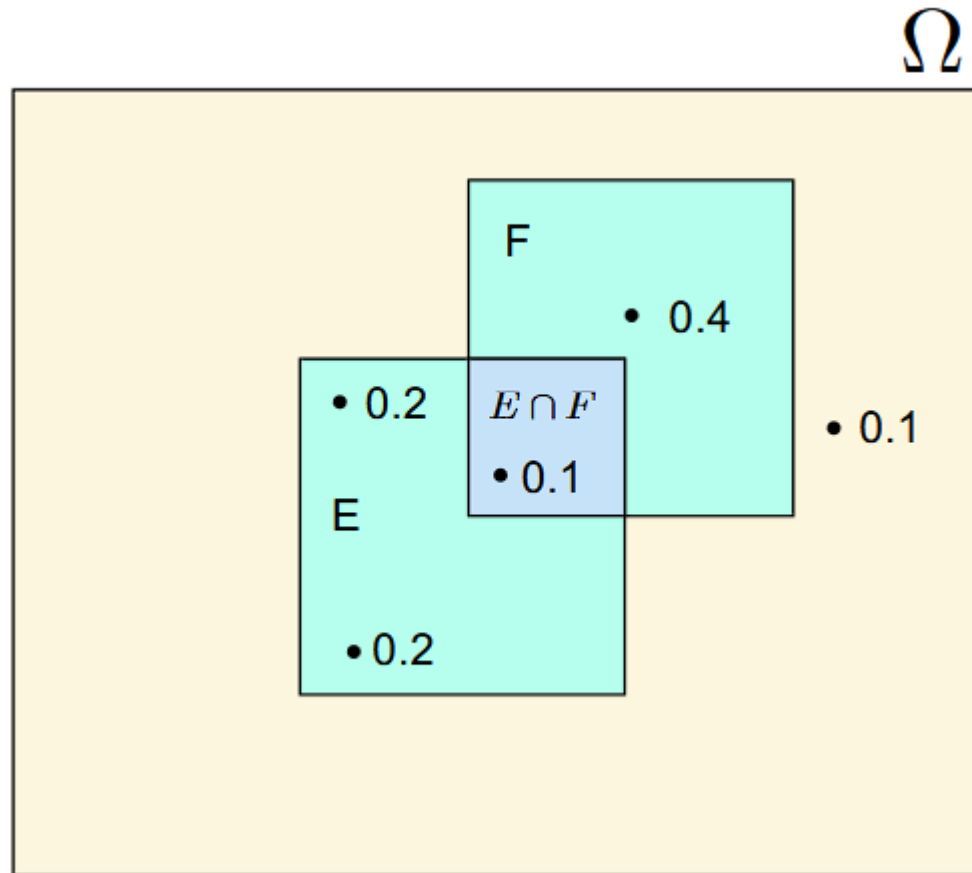
$$P(\bar{E}) = 1 - P(E)$$

Odds

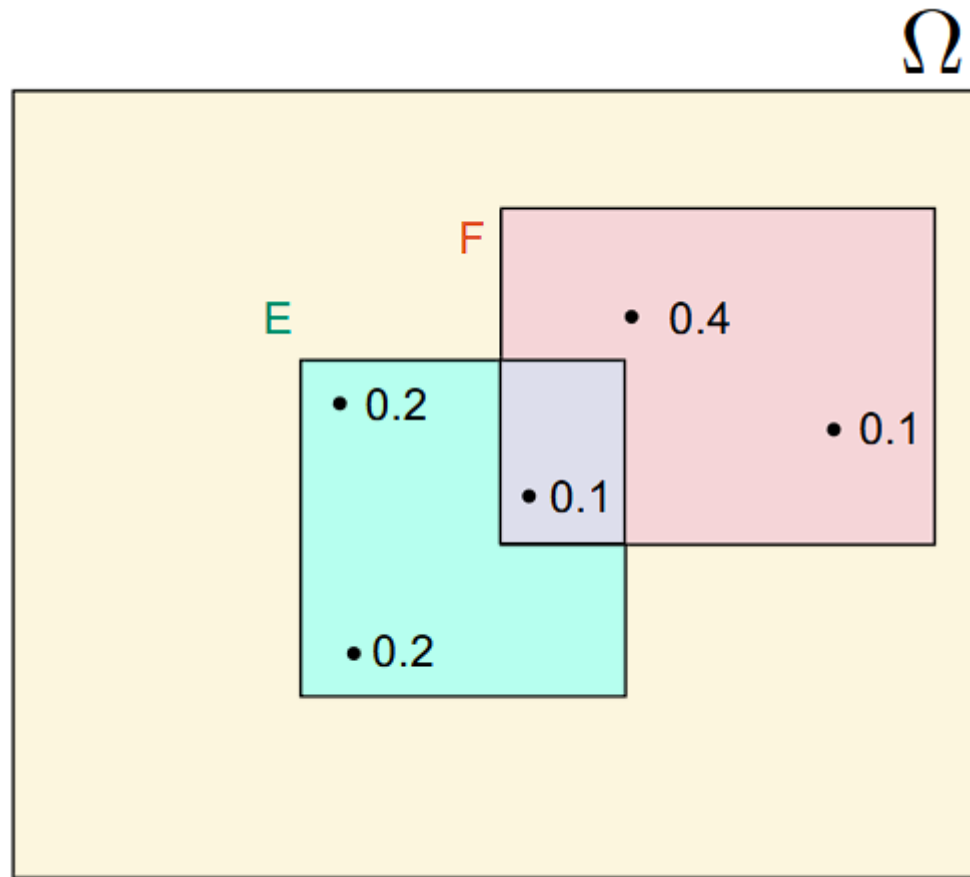


$$O_P(E) = \frac{P(E)}{P(\bar{E})}$$

Intersection & Union

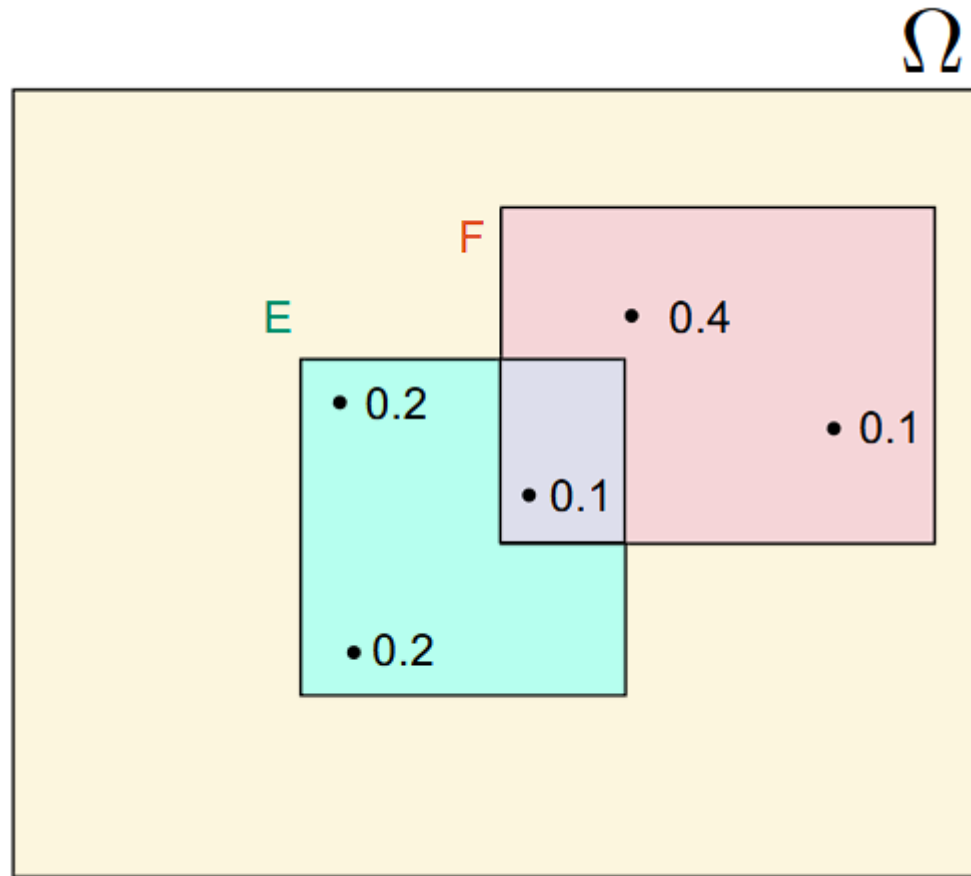


Partition Rule



$$P(E) = P(E \cap F) + P(E \cap \bar{F})$$

Conditional Probability



$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

Formulas

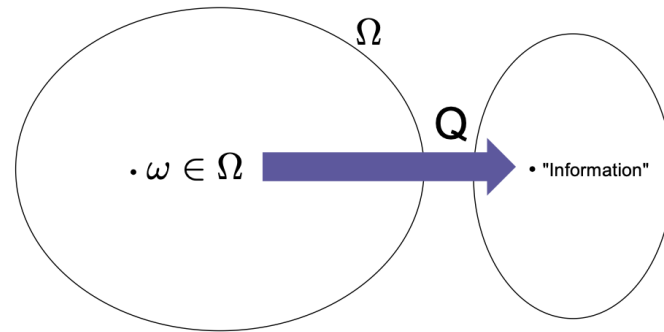
- Chain Rule: $P(E \cap F) = P(E|F) * P(F)$
- Independence: $P(E|F) = P(E)$ or equivalently $P(E \cap F) = P(E) * P(F)$
- Bayes' Rule: $P(E|F) = \frac{P(F|E)*P(E)}{P(F)}$
 - Evidence: $P(F)$
 - Prior: $P(E)$
 - Posterior: $P(E|F)$

Random Variables

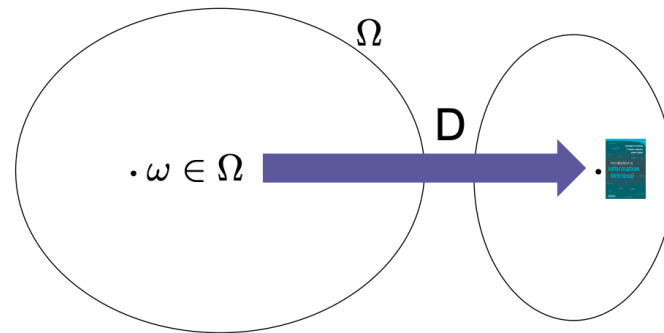
- Random Variable: $(X: \Omega \rightarrow \mathbb{R})$
 - not random nor a variable
 - Function mapping elementary events of Ω to $\mathbb{R}$
- **Notation:**
 - $P_X(x) = \dots$ or equivalently $P(X = x) = \dots$
 - $P_{XY}(x, y) = \dots$ or equivalently $P(X = x, Y = y) = \dots$
 - $P_{X|Y}(x, y) = \dots$ or equivalently $P(X = x | Y = y) = \dots$

Probability Model

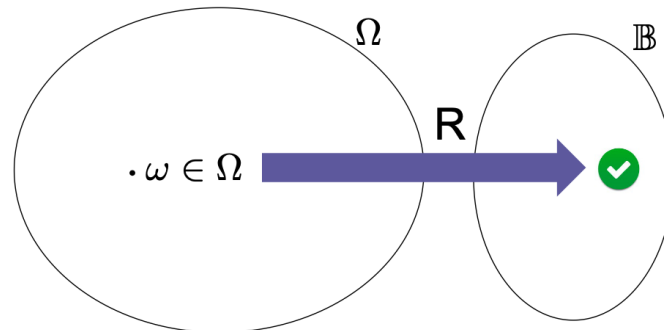
- Query as a random variable:



- Document as a random variable



- Relevance as a random variable



Probability Model

- Now we can model relevance as: $P(R = 1 \mid D = d \wedge Q = q)$
- Retrieve documents if:
 - $P(R = 1 \mid D = d \wedge Q = q) > P(R = 0 \mid D = d \wedge Q = q)$
- Ranking: sort by probability of relevance (**Probability Ranking Principle**)

Binary Independence Model

- Treat documents and queries as Boolean vectors:

$$P(D_k = d_k | R = 1 \wedge Q = q)$$

If $d_k = 1$: Probability that document **d** contains term **k**, knowing that the query is **q** and that **d** is relevant to the information need behind **q**.

If $d_k = 0$: Probability that document **d** does not contain term **k**, knowing that the query is **q** and that **d** is relevant to the information need behind **q**.

Binary Independence Model

- Naïve Bayes Assumption:

$$P(D = d) = \prod_{k=1}^{k=M} P(D_k = d_k)$$

- i.e. we assume terms in a document occur independently of each other

Deriving the RSV

- We want to obtain some probabilistic score for document relevance.
- **Step 1: Bayes' Formula**

$$P(R = 1|D = d) = \frac{P(D = d|R = 1) \times P(R = 1)}{P(D = d)}$$

- **Step 2: Condition on query q**

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

Deriving the RSV

- **Step 3: Use odds to get rid of the denominator**

$$P(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|Q = q)}$$

$$P(R = 0|D = d \wedge Q = q) = \frac{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}{P(D = d|Q = q)}$$



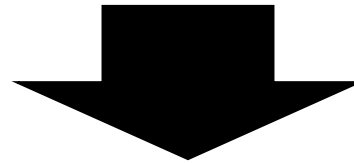
$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q) \times P(R = 1|Q = q)}{P(D = d|R = 0 \wedge Q = q) \times P(R = 0|Q = q)}$$

Deriving the RSV

- **Step 4: Simplify**

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times \frac{P(R = 1|Q = q)}{P(R = 0|Q = q)}$$

These are odds!



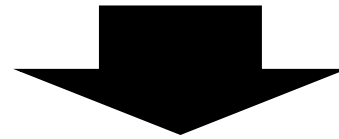
$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

These are odds!

Deriving the RSV

- **Step 5: Use the independence assumption**

$$O(R = 1|D = d \wedge Q = q) = \frac{P(D = d|R = 1 \wedge Q = q)}{P(D = d|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$



$$O(R = 1|D = d \wedge Q = q) = \frac{\prod_{k=1}^{k=M} P(D_k = d_k|R = 1 \wedge Q = q)}{\prod_{k=1}^{k=M} P(D_k = d_k|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Deriving the RSV

- **Step 6: Split the probabilities**

$$O(R = 1|D = d \wedge Q = q) = \prod_{k=1}^{k=M} \frac{P(D_k = d_k | R = 1 \wedge Q = q)}{P(D_k = d_k | R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

This is boolean!



$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=0} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{k|d_k=1} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

Deriving the RSV

- **Step 7: Introduce two new variables**

$$O(R = 1|D = d \wedge Q = q) = \prod_{\substack{k|d_k=0 \\ \wedge q_k = 1}} \frac{P(D_k = 0|R = 1 \wedge Q = q)}{P(D_k = 0|R = 0 \wedge Q = q)} \times \prod_{\substack{k|d_k=1 \\ \wedge q_k = 1}} \frac{P(D_k = 1|R = 1 \wedge Q = q)}{P(D_k = 1|R = 0 \wedge Q = q)} \times O(R = 1|Q = q)$$

We call this p_k

We call this u_k

	kth term present	kth term absent
Relevant	p_k	$1-p_k$
Not relevant	u_k	$1-u_k$

Deriving the RSV

- **Step 8: Divide and multiply with the same product**

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k}{u_k} \times \prod_{k|d_k=0 \wedge q_k=1} \frac{1-p_k}{1-u_k} \times O(R = 1|Q = q)$$

divide with $\prod_{k|d_k=1 \wedge q_k=1} \frac{1-p_k}{1-u_k}$

multiply with $\prod_{k|d_k=1 \wedge q_k=1} \frac{1-p_k}{1-u_k}$

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1-u_k)}{u_k \times (1-p_k)} \times \prod_{k|q_k=1} \frac{1-p_k}{1-u_k} \times O(R = 1|Q = q)$$

Deriving the RSV

- **Step 9: Get rid of the constant**

$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)} \times \prod_{k|q_k=1} \frac{1 - p_k}{1 - u_k} \times O(R = 1|Q = q)$$

This does not depend on d!



$$O(R = 1|D = d \wedge Q = q) = \prod_{k|d_k=1 \wedge q_k=1} \frac{p_k \times (1 - u_k)}{u_k \times (1 - p_k)}$$

Deriving the RSV

- **Step 10: Rewrite and take log**

$$RSV_d = \log \prod_{k|d_k=1 \wedge q_k=1} \frac{\frac{p_k}{1-p_k}}{\frac{u_k}{1-u_k}}$$

Deriving the RSV

- **Step 11: Use some magical log formulas**

$$RSV_d = \sum_{k | d_k=1 \wedge q_k=1} \log \boxed{\frac{p_k}{1-p_k}} - \log \boxed{\frac{u_k}{1-u_k}}$$

Odds of containing term k
in relevant documents

Odds of containing term k
in non-relevant documents

	k th term present	k th term absent
Relevant	p_k	$1-p_k$
Not relevant	u_k	$1-u_k$

Deriving the RSV

- **Step 12: Introduce a new variable**

$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{p_k}{1-p_k} - \log \frac{u_k}{1-u_k}$$

c_k



$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} c_k$$

Estimating Weights

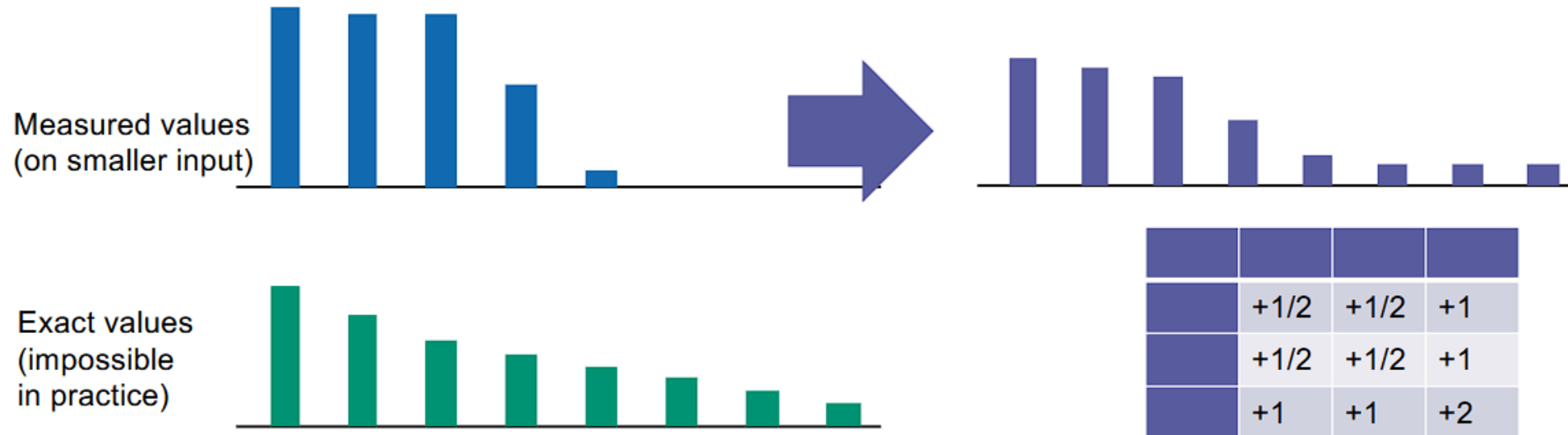
- N – total number of documents
- S – total number of relevant documents
- df_k – number of documents containing term k
- s – number of relevant documents containing term k

	k^{th} term present	k^{th} term absent	Total
Relevant	s	$S-s$	S
Not relevant	$df_k - s$	$N - df_k - S + s$	$N-S$
Total	df_k	$N - df_k$	N

Smoothing

- We want to avoid division and log by 0 and prevent extreme weights caused by low counts in the log

	k^{th} term present	k^{th} term absent
Relevant	s_k +1/2	$S_k - s_k$ +1/2
Not relevant	$df_k - s_k$ +1/2	$N - df_k - S_k + s_k$ +1/2



Estimating Weights

- In practice we don't know S and s . Croft and Harper suggest that odds are constant. So, we get:

	k^{th} term present	k^{th} term absent
Relevant	s_k	$S_k - s_k$
Not relevant	$df_k - s_k$	$N - df_k - S_k + s_k$



	k^{th} term present	k^{th} term absent
Relevant	$1/2$	$1/2$
Not relevant	$df_k - s$	$N - df_k - S + s$

Estimating Weights

- Most documents are not relevant ($s, S \ll df_k, N$), so we get:

	k^{th} term present	k^{th} term absent
Relevant	1/2	1/2
Not relevant	$df_k - s$	$N - df_k - S + s$



	k^{th} term present	k^{th} term absent
Relevant	1/2	1/2
Not relevant	df_k	$N - df_k$

Estimating Weights

- With smoothing:

	k^{th} term present	k^{th} term absent
Relevant	$1/2$	$1/2$
Not relevant	df_k	$N - df_k$



	k^{th} term present	k^{th} term absent
Relevant	$1/2$	$1/2$
Not relevant	df_k	N

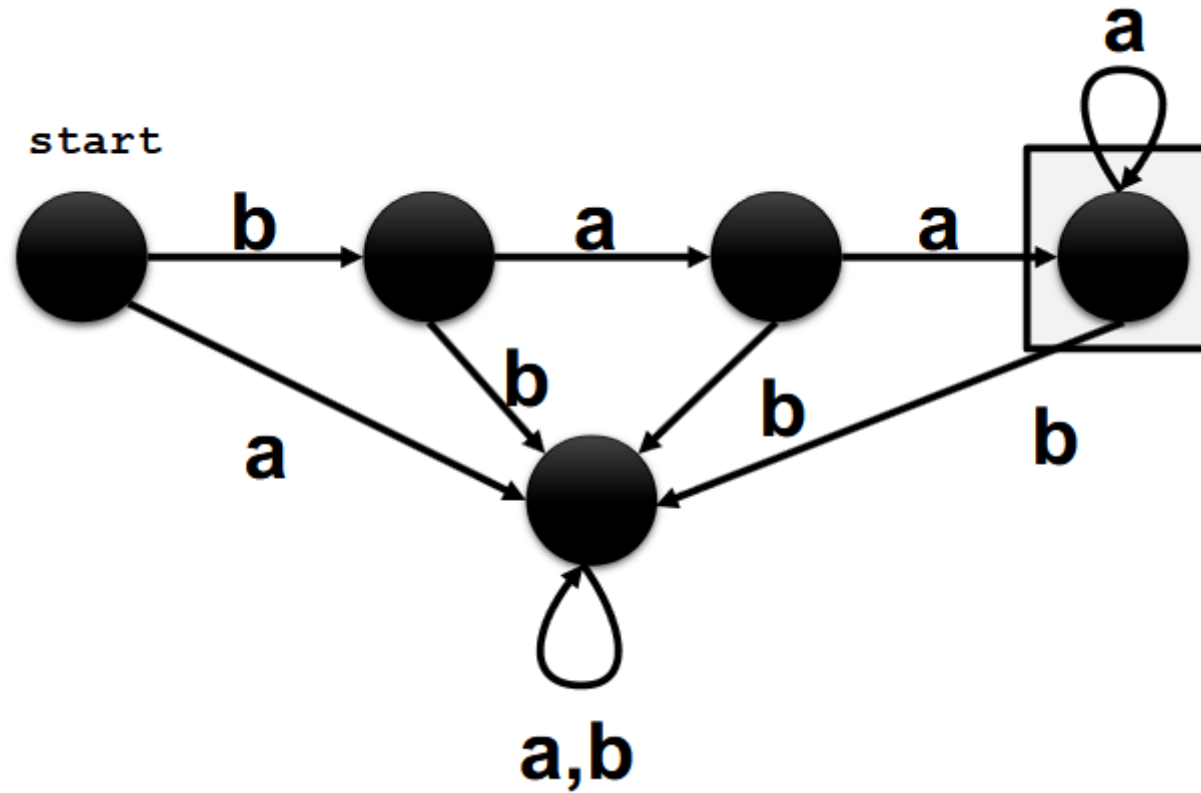
Estimating Weights

- With this, the weights just become the inverted document frequency

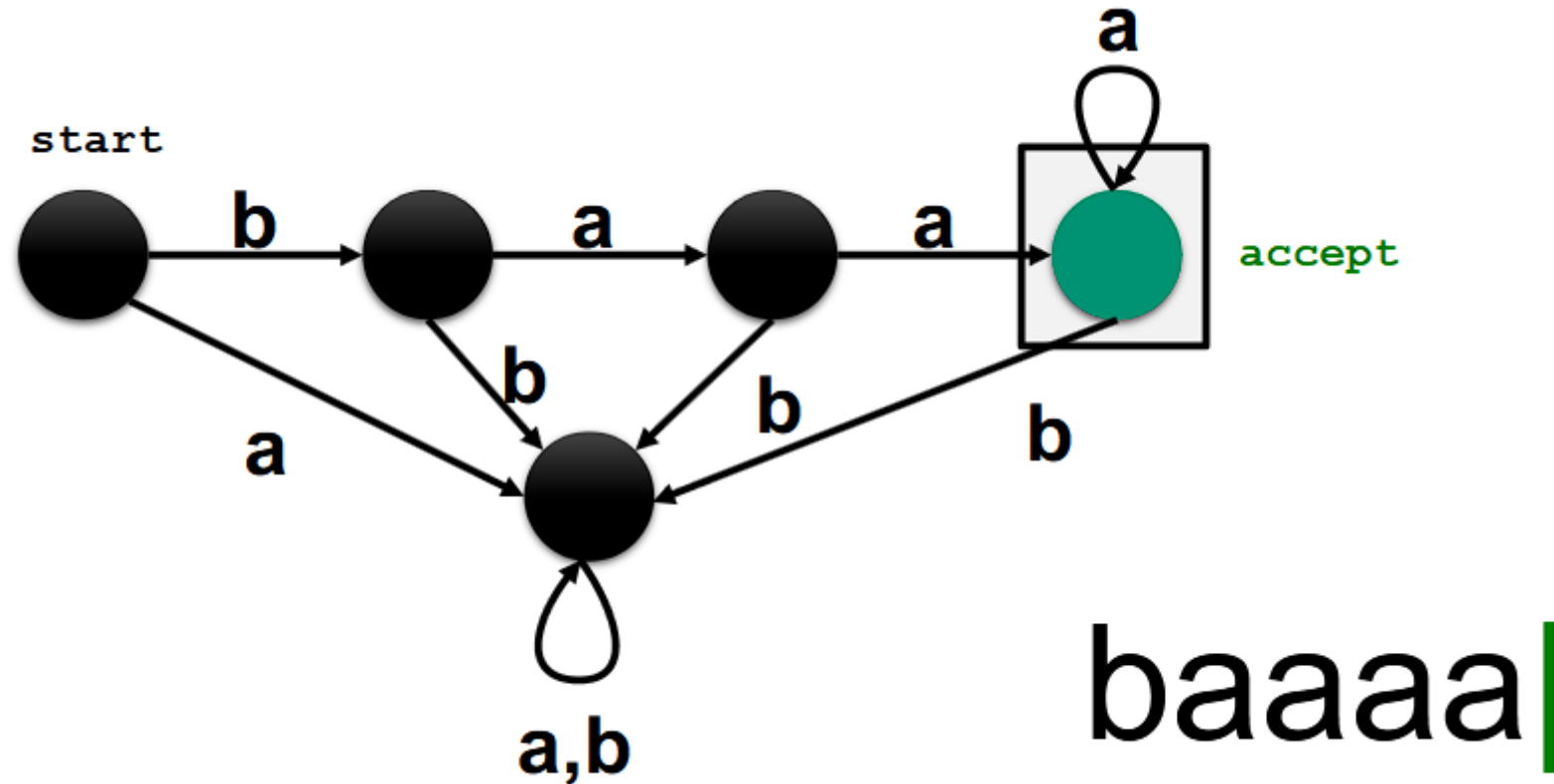
$$RSV_d = \sum_{k|d_k=1 \wedge q_k=1} \log \frac{N}{df_k}$$

That seems like a lot of effort just to get the same result as before. But Croft and Harpers assumptions are heavily debated. Also, we can improve the odds by iteratively taking in user feedback (Relevance Feedback), this is the main difference to the VSM.

Language Models

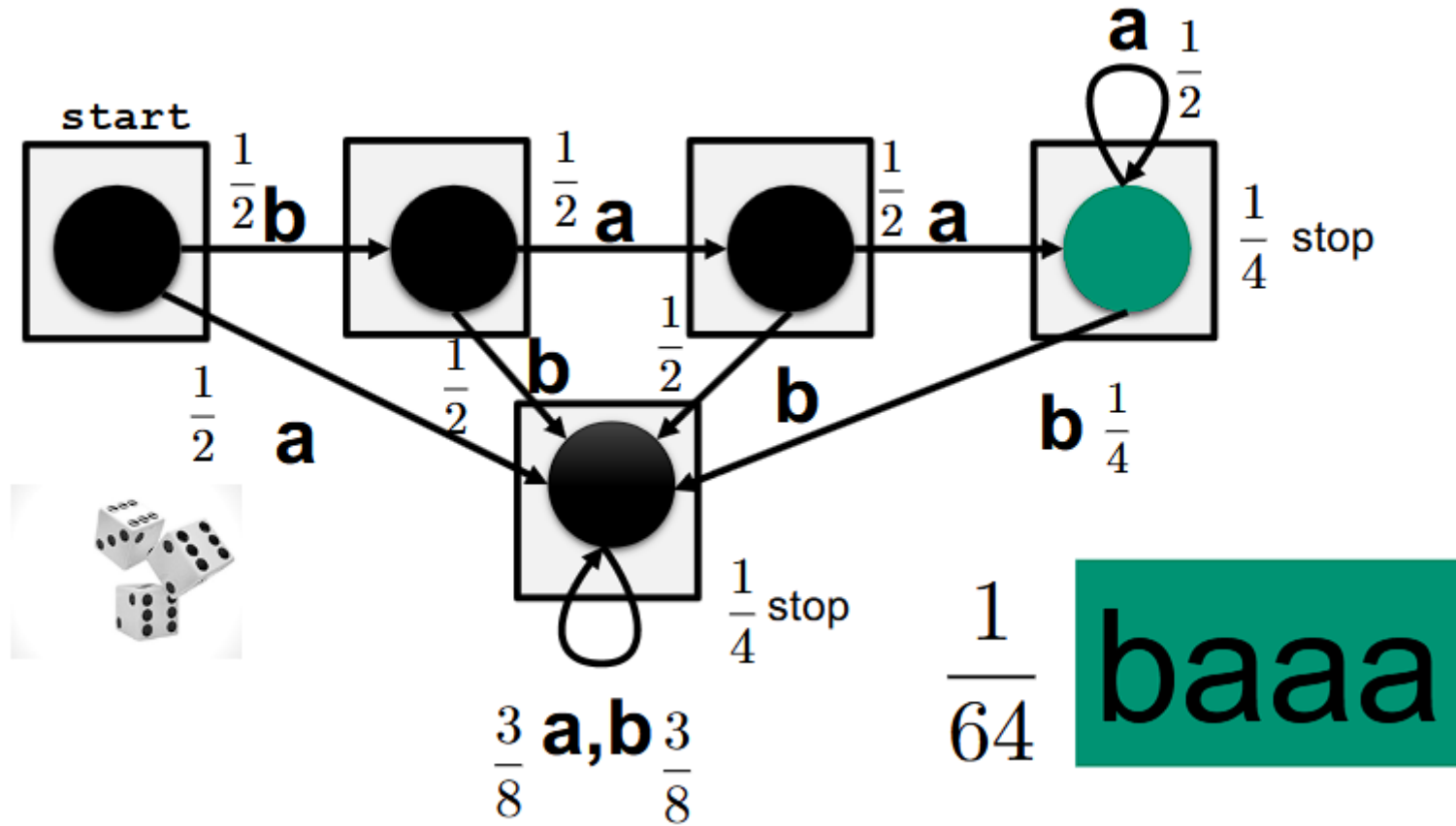


Finite State Automaton (FSA)



Finite State Automaton (FSA)

- Turn it into a generator:



Markov rule

- (n+1)-gram:

$$P(D_k = d_k | L_D \geq k \wedge D_1 = d_1 \wedge D_2 = d_2 \wedge \dots \wedge D_{k-1} = d_{k-1})$$

=

$$P(D_k = d_k | L_D \geq k \wedge D_{k-n} = d_{k-n} \wedge \dots \wedge D_{k-1} = d_{k-1})$$

Only look at the previous n terms!

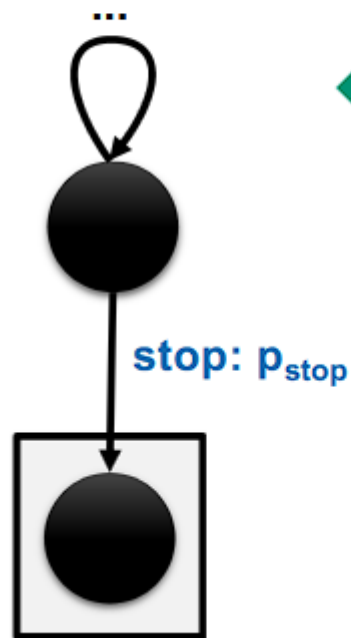
Markov rule

- Example: document of length 4

$$\begin{aligned} P(D = (d_1, d_2, d_3, d_4)) = & \\ & (1 - p_{stop})P(D_1 = d_1 | L_D \geq 1) \\ & (1 - p_{stop})P(D_2 = d_2 | D_1 = d_1 \wedge L_D \geq 2) \\ & (1 - p_{stop})P(D_3 = d_3 | D_1 = d_1 \wedge D_2 = d_2 \wedge L_D \geq 3) \\ & (1 - p_{stop})P(D_4 = d_4 | D_2 = d_2 \wedge D_3 = d_3 \wedge L_D \geq 4)p_{stop} \end{aligned}$$

Unigram Model

$1 - p_{\text{stop}}$
to: 0.3
be: 0.2
or: 0.1
not: 0.05
...



t	p_t
to	0.3
be	0.2
or	0.1
not	0.05
that	0.15
is	0.1

Unigram model

Bigram Model: Conditioned on previous term

Kahoot

<https://create.kahoot.it/share/incomplete-information-retrieval-probabilistic-retrieval/d01d9bd5-1220-468b-a793-2e5f1146d8e2>